

SIMATS ENGINEERING

ASSESSMENT TOOL I– INDUSTRY PROBLEM TASK

Course: Ethical Hacking

Code: ITA1407

Total Marks: 100

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COURSE OUTCOME COVERED

CO5: Analyze vulnerabilities in Windows and Linux systems and evaluate security weaknesses in messaging, web, and mobile applications using appropriate exploitation techniques and hacking tools. (BT5)

1. A mid-sized financial organization operates a Windows domain network consisting of 150 interconnected systems. During a security incident investigation, it is observed that attackers have exploited NULL sessions, open RPC services, and misconfigured shared folders to gain unauthorized access. Initial analysis indicates that 45% of systems allow anonymous enumeration, and among those, 60% are susceptible to SMB-based lateral movement attacks. Furthermore, each compromised system can attempt to access 3 additional systems, with a success probability of 0.3 per attempt.

Question:

Analyze the extent of the attack by calculating the number of systems compromised during the initial stage of exploitation. Further, estimate the spread of the attack in the second stage due to lateral movement. Based on your analysis, evaluate whether the network is at risk of a large-scale compromise (greater than 70% of total systems), and justify which vulnerability (NULL sessions, RPC exposure, or weak shares) should be prioritized for immediate mitigation.

COMANDS AND OUTPUT:

1. `nmap -sn 192.168.1.0/24`

```
└─$ nmap -sn 192.168.1.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 01:37 -0400
Nmap scan report for 192.168.1.0
Host is up (0.00068s latency).
Nmap scan report for 192.168.1.1
Host is up (0.024s latency).
Nmap scan report for 192.168.1.2
Host is up (0.00022s latency).
Nmap scan report for 192.168.1.3
Host is up (0.00022s latency).
Nmap scan report for 192.168.1.251
Host is up (0.0014s latency).
Nmap scan report for 192.168.1.252
Host is up (0.00053s latency).
Nmap scan report for 192.168.1.253
Host is up (0.00022s latency).
Nmap scan report for 192.168.1.254
Host is up (0.00057s latency).
Nmap scan report for 192.168.1.255
Host is up (0.00052s latency).
Nmap done: 256 IP addresses (256 hosts up) scanned in 53.69 seconds
```

2. `nmap -p 135,139,445 192.168.1.0/24`

```
(kali@kali)-[~]
└─$ nmap -p 135,139,445
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 01:38 -0400
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.05 seconds
```

3. `nmap -p 445 --script smb-enum-shares 192.168.1.254`

```
(kali@kali)-[~]
└─$ nmap -p 135,139,445 192.168.1.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 01:38 -0400
Nmap scan report for 192.168.1.0
Host is up (0.0017s latency).

PORT      STATE      SERVICE
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds

Nmap scan report for 192.168.1.1
Host is up (0.011s latency).

PORT      STATE      SERVICE
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds

Nmap scan report for 192.168.1.2
Host is up (0.026s latency).

PORT      STATE      SERVICE
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds

Nmap scan report for 192.168.1.3
Host is up (0.0013s latency).

PORT      STATE      SERVICE
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds

Nmap scan report for 192.168.1.4
Host is up (0.00032s latency).

PORT      STATE      SERVICE
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds

Nmap scan report for 192.168.1.5
Host is up (0.025s latency).
```

4. `nmap -p 445 --script smb-enum-users 192.168.1.254`

```
(kali@kali)-[~]
$ nmap -p 445 --script smb-enum-shares,smb-enum-users 192.168.1.254
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 01:42 -0400
Nmap scan report for 192.168.1.254
Host is up (0.00070s latency).

PORT      STATE      SERVICE
445/tcp    filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 2.45 seconds
```

5. `nmap -p 445 --script smb-protocols 192.168.1.254`

```
(kali@kali)-[~]
$ nmap -p 445 --script smb-enum-users 192.168.1.254
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 01:43 -0400
Nmap scan report for 192.168.1.254
Host is up (0.00037s latency).

PORT      STATE      SERVICE
445/tcp    filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 2.33 seconds
```

6. `nmap -p 139 --script nbstat 192.168.1.254`

```
(kali@kali)-[~]
$ nmap -p 139 --script nbstat 192.168.1.254
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 01:45 -0400
Nmap scan report for 192.168.1.254
Host is up (0.0021s latency).

PORT      STATE      SERVICE
139/tcp    filtered  netbios-ssn

Nmap done: 1 IP address (1 host up) scanned in 2.43 seconds
```

7. `nmap -p 445 --script smb-vuln* 192.168.1.254`

```
(kali@kali)-[~]
$ nmap -p 445 --script smb-vuln* 192.168.1.254
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 01:45 -0400
Nmap scan report for 192.168.1.254
Host is up (0.00074s latency).

PORT      STATE      SERVICE
445/tcp    filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 2.43 seconds
```

2. An enterprise network consists of 120 systems, out of which 70 systems expose RPC services to the internal network. A vulnerability scan reveals that 40% of these systems are susceptible to known exploits, and the probability of successful exploitation for each vulnerable system is 0.5. However, each exploitation attempt carries a detection probability of 0.25 due to intrusion detection systems deployed within the network.

Question:

Determine the expected number of systems that can be successfully compromised by an attacker. Additionally, compute the expected number of detection alerts generated during the exploitation process. Based on your results, critically evaluate whether an attacker aiming to remain stealthy would proceed with exploitation or adopt alternative attack strategies, providing justification for your reasoning.

COMANDS AND OUTPUT:

1. `nmap -p 135 192.168.1.254`

```
(kali㉿kali)-[~]
$ nmap -p 135 192.168.1.254
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 01:58 -0400
Nmap scan report for 192.168.1.254
Host is up (0.00096s latency).

PORT      STATE      SERVICE
135/tcp    filtered  msrpc

Nmap done: 1 IP address (1 host up) scanned in 2.33 seconds
```

2. `nmap -p 135 --script rpcinfo 192.168.1.254`

```
(kali㉿kali)-[~]
$ nmap -p 135 --script rpcinfo 192.168.1.254
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 01:59 -0400
Nmap scan report for 192.168.1.254
Host is up (0.0012s latency).

PORT      STATE      SERVICE
135/tcp    filtered  msrpc

Nmap done: 1 IP address (1 host up) scanned in 2.39 seconds
```

3. nmap -sV -p 111,135,139,445 192.168.1.254

```
(kali@kali)-[~]
$ nmap -sV -p 111,135,139,445 192.168.1.254
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 02:00 -0400
Nmap scan report for 192.168.1.254
Host is up (0.00063s latency).

PORT      STATE      SERVICE      VERSION
111/tcp    filtered  rpcbind
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.58 seconds
```

4. nmap -Pn -p 135,445 192.168.1.254

```
(kali@kali)-[~]
$ nmap -Pn -p 135,445 192.168.1.254
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 02:00 -0400
Nmap scan report for 192.168.1.254
Host is up (2.0s latency).

PORT      STATE      SERVICE
135/tcp    filtered  msrpc
445/tcp    closed    microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 21.18 seconds
```

3. A cloud-hosted Linux server is running multiple services exposed to the internet. Security analysis identifies four critical services, with independent probabilities of vulnerability being 0.3, 0.2, 0.25, and 0.15, respectively. It is also observed that the system has weak file permissions, which can be exploited with a probability of 0.5 once initial access is gained.

Question:

Evaluate the probability that at least one service is vulnerable and can be exploited by an attacker. Further, compute the overall probability of a full system compromise, considering both service exploitation and privilege escalation through weak permissions. Based on your findings, analyze whether strengthening file permissions alone would significantly reduce the overall risk, and justify your conclusion.

COMANDS AND OUTPUT:

1. nmap -sV 192.168.1.254

```
(kali@kali)-[~]
$ nmap -sV 192.168.1.254
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 02:05 -0400
Nmap scan report for 192.168.1.254
Host is up (0.0021s latency).
All 1000 scanned ports on 192.168.1.254 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.96 seconds
```

2. nmap -p- 192.168.1.254

```
(kali㉿kali)-[~]  
$ nmap -p- 192.168.1.254  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 02:07 -0400  
Stats: 0:01:04 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 22.83% done; ETC: 02:11 (0:03:33 remaining)  
Stats: 0:02:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 46.94% done; ETC: 02:11 (0:02:21 remaining)  
Stats: 0:03:30 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 80.42% done; ETC: 02:11 (0:00:51 remaining)  
Nmap scan report for 192.168.1.254  
Host is up (0.042s latency).  
All 65535 scanned ports on 192.168.1.254 are in ignored states.  
Not shown: 65535 filtered tcp ports (no-response)  
  
Nmap done: 1 IP address (1 host up) scanned in 260.77 seconds
```

3. nmap -sV -sC 192.168.1.254

```
(kali㉿kali)-[~]  
$ nmap -sC -sC 192.168.1.254  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 02:12 -0400  
Nmap scan report for 192.168.1.254  
Host is up (0.0013s latency).  
All 1000 scanned ports on 192.168.1.254 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Nmap done: 1 IP address (1 host up) scanned in 12.21 seconds
```

4. nmap -p 80,443 --script http-title,http-headers 192.168.1.254

```
(kali㉿kali)-[~]  
$ nmap -p 80,443 --script http-title,http-headers 192.168.1.254  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 02:14 -0400  
Nmap scan report for 192.168.1.254  
Host is up (0.0011s latency).  
  
PORT      STATE      SERVICE  
80/tcp    filtered  http  
443/tcp   filtered  https  
  
Nmap done: 1 IP address (1 host up) scanned in 3.39 seconds
```

4. An organization uses Network File System (NFS) to share resources across three internal subnets. Due to improper configuration, unauthorized access probabilities from these subnets are estimated as 0.4, 0.3, and 0.2, respectively. Once access to the NFS share is obtained, the probability of privilege escalation to root access is 0.7.

Question:

Calculate the probability that at least one subnet successfully gains unauthorized access to the NFS share. Subsequently, determine the overall probability of a complete system compromise. Based on your analysis, evaluate the effectiveness of restricting NFS access to a single subnet as a mitigation strategy, and justify your recommendation.

COMANDS AND OUTPUT:

1. `nmap -p 111,2049 192.168.1.254`

```
(kali㉿kali)-[~]
$ nmap -p 111,2049 192.168.1.254
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 02:17 -0400
Nmap scan report for 192.168.1.254
Host is up (0.00062s latency).

PORT      STATE      SERVICE
111/tcp    filtered  rpcbind
2049/tcp   filtered  nfs

Nmap done: 1 IP address (1 host up) scanned in 3.32 seconds
```

2. `nmap -p 111,2049 --script nfs-showmount 192.168.1.254`

```
(kali㉿kali)-[~]
$ nmap -p 111,2049 --script nfs-showmount 192.168.1.254
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-25 02:18 -0400
Nmap scan report for 192.168.1.254
Host is up (0.00060s latency).

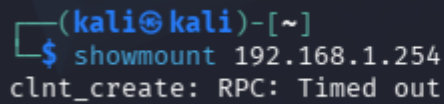
PORT      STATE      SERVICE
111/tcp    filtered  rpcbind
2049/tcp   filtered  nfs

Nmap done: 1 IP address (1 host up) scanned in 3.37 seconds
```

3. `rpcinfo -p 192.168.1.254`

```
(kali㉿kali)-[~]
$ rpcinfo -p 192.168.1.254
192.168.1.254: RPC: Remote system error - Connection refused
```


4. `showmount -e 192.168.1.254`



```
(kali㉿kali)-[~]  
$ showmount 192.168.1.254  
clnt_create: RPC: Timed out
```

5. An e-commerce platform maintains a database containing 50,000 user records, of which 30% are classified as sensitive. A successful SQL injection attack allows data extraction at a rate of 500 records per second. However, the application firewall detects and blocks the attack

after 40 seconds.

Question:

Compute the total number of records that can be extracted before the attack is blocked.

Determine the percentage of total data compromised, and estimate the number of sensitive records exposed. Based on your analysis, evaluate the severity of the breach and justify whether the existing detection mechanism is sufficient to protect critical data.

Answer:

Given:

- Total records = 50,000
- Sensitive records = 30%
- Extraction rate = 500 records/second
- Detection time = 40 seconds

Calculations

1. Records extracted

$$500 \times 40 = 20,000 \text{ records}$$

2. Percentage of database compromised

$$(20,000 / 50,000) \times 100 = 40\%$$

3. Sensitive records exposed

$$20,000 \times 30\% = 6,000 \text{ records}$$

Final answer

- Records extracted: 20,000
- Total data compromised: 40%
- Sensitive records exposed: 6,000
- Severity: High, because 40% of the entire database can be extracted before detection.

